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TRIZ as an innovation tool for opportunity management and Lean Construction

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Abstract

This paper explores the integration of the Theory of Inventive Problem-Solving (TRIZ) into project management practices, particularly alongside opportunity management and Lean Construction methodologies. A survey with almost 100 practitioners showed that the use of TRIZ is non-existent in the Norwegian construction industry. Through a literature review, the paper investigates how TRIZ enhances innovation efforts. Core TRIZ theory is presented, and the applicability of the 40 Inventive Principles tool is used to illustrate how TRIZ can be used in the innovative process of identifying opportunities, minimizing waste, and enhancing value within projects. Additionally, we highlight synergies between TRIZ, opportunity management, and Lean Construction, emphasizing their collective role in improving project outcomes. This study contributes to advancing theoretical insights and practical approaches for innovation and improved project success in construction projects.

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1. Introduction

The construction industry is experiencing a change in approach due to increased complexity. *Project management success* is often evaluated in terms of the iron triangle of cost, time, and quality [1]. The iron triangle is especially on the agenda during the execution phase of a project, making it common for practitioners to drift back to a mindset where customer value is forgotten, and the purpose changes back to delivering on the predefined frames of time, cost, and quality. However, there is an understanding between scholars that *project success* also requires a valuable end product for the customer or user [1-3]. These issues are getting more on the agenda, along with the increased global attention to sustainability.

When projects are conducting uncertainty management efforts, the mind is usually just set on the negative uncertainty or threats [4, 5]. This mindset can be problematic because it gives no room for improvement. If all your efforts are to keep the project within the predefined frames by only dealing with potential threats while neglecting the opportunities, it is essentially impossible to perform *better* than the predefined frames [6]. However, suppose you balance the uncertainty management effort and try to identify and implement opportunities. In that case, you may get a better end result than what was previously defined at an early stage.

A major reason practitioners return to this threat-based mindset is that there are no real incentives for the project practitioners to keep looking for opportunities to improve the project. With Lean Construction, there has been an effort to create incentives to continuously improve the project to the better, with customer value as the focal point. Malvik et al. showed that the Lean Construction approach Target Value Delivery (TVD) and opportunity management can work as complementary practices [7]. With TVD, the project practitioners are incentivized to look for opportunities for improvement throughout the project, including the execution phase. In TVD, customer value is vital in achieving project success rather than project management success.

Common for both opportunity management and Lean Construction are the continuous efforts to look for ways to improve a project's end result and secure better long-term effects. However, as research has shown, it is often neglected, especially in the execution phase [8]. Innovation is an important factor for improvement. A barrier to implementing effective opportunity management in a project might be the lack of structured ways to innovate.

The Theory of Inventive Problem-Solving (TRIZ) is a philosophy that uses problem-solving tools and methods created to develop creativity within groups. A vital point of TRIZ is to show how all innovation is just small adjustments of inventions that already exist [9]. The TRIZ philosophy can make the innovation process in opportunity management or Lean Construction easier by offering techniques for finding innovation in different situations. The main author conducted a survey with 94 practitioners in the Norwegian construction industry. The survey results showed that none of the 94 respondents had heard about or used TRIZ in their project management efforts.

This paper looks into the potential of using TRIZ for innovation in project management efforts by investigating how it can complement opportunity management and Lean Construction. Thus, the research question is:

How can TRIZ work as a tool to enhance innovation efforts in opportunity management and Lean Construction?

The following sections of the paper will start with the method first before providing a theoretical background of TRIZ, opportunity management, and Lean Construction. Then, there is a discussion section where the three topics are seen in relation to each other before finishing off with the conclusion.

2. Method

A literature review was undertaken to gather quantitative data involving searches in books, journals, and papers accessible through university libraries or web search engines. The literature centered around opportunity management and Lean Construction has been gathered over several years of research. The literature about TRIZ was done more specifically for this study and involved searching for relevant literature in web search engines. Snowballing was used to understand TRIZ as a concept through the most relevant primary sources of the theory. The paper is conceptual, as the ideas have not been tested in any project, and the data are primarily theoretical. Jaakkola [10] suggests four common research designs for conceptual papers: theory synthesis, theory adaption,

typology, and model. This paper uses the *model* approach, where a relationship between different constructs is predicted, and the purpose of the research is to analyze and explain how these constructs are related based on theoretical insight.

The author conducted a large survey with practitioners from various roles within the Norwegian construction industry. Although the survey was originally designed for other research purposes outside the scope of this paper, the author included one specific question about the use of TRIZ in their projects. In this question, respondents were given a list of 11 tools associated with Lean methodologies and were asked to mark each tool that had been used in their projects. The responses ranged from 0 to 59 out of the 94 participants. Notably, TRIZ was the only tool that received zero checks.

Of the respondents, about 75 percent represented clients, 20 percent represented consultants or contractors, and five percent represented owners. The survey was presented at a project management seminar. In addition, key personnel in many of the biggest actors in the Norwegian construction industry were contacted through e-mail, and they forwarded the survey to relevant colleagues. Microsoft Forms was used to collect the responses, which were then analyzed and sorted using its integrated sorting features before being exported to Microsoft Excel for further analysis. However, minimal analysis was required for this research paper, as the results clearly supported the hypothesis that TRIZ is rarely used, evidenced by its zero responses.

3. Theory

3.1. Theory of Inventive Problem-Solving (TRIZ)

The Theory of Inventive Problem-Solving (TRIZ) is a systematic approach to problem-solving. The TRIZ concept goes all the way back to 1948 when Genrik Altshuller developed it as an attempt to theorize creativity [11]. The effort was based on an enormous dataset of over 400 000 technology patents [12]. The TRIZ philosophy is now based on a study of over two million successful patents, according to Mann [9]. He argues that the idea with TRIZ is not to replace creativity but rather to offer a structure to people's innovation. Retseptor argues that the forty patterns of inventive solutions discovered by Altshuller, the 40 Inventive Principles, apply to both technical and non-technical fields. He argues that the 40 Inventive Principles are fundamental, universal, and powerful instruments of human creativity and presents several examples of how the toolkit can be used in Quality Management [13]. While TRIZ originally was focused on technology patents, the approach is applicable to all types of fields. According to Mann [9], the most vital findings of TRIZ are that:

- A limited number of inventive principles and strategies drive the emergence of all kinds of innovation.
- Trends in technological evolution are often highly predictable.
- The best solutions transform unwanted or harmful elements into valuable resources.
- Effective solutions actively identify and eliminate conflicts and trade-offs that are traditionally considered fundamental in design practices.

According to TRIZ, innovation is rooted in universal creativity principles, often involving adapting theories from different fields. Essentially, most problems have likely been addressed in some form elsewhere, although with different perspectives and in diverse contexts. Solutions to your current challenge may already exist, albeit in different forms. Creative problem-solving entails discovering and adapting these existing solutions to your specific needs. Recognizing this reality can alleviate the pressure of innovation, underscoring the notion that reinventing the wheel is unnecessary for project innovation. While TRIZ may seem complex, the general TRIZ process is simple. You take your specific problem, fit it into a generic problem already defined in TRIZ, and look at the generic solution provided by TRIZ before finally adjusting it to fit your specific solution.

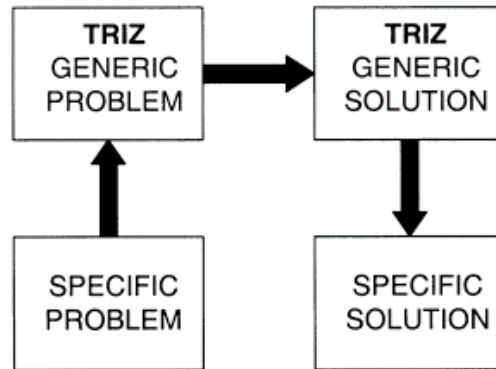


Fig. 1. The TRIZ process (retrieved from Mann [9])

A tool that potentially can be used for this simple process is the 40 Inventive Principles. Retseptor [13] showed how the tool could be used in an approach to Quality Management, and Navas et al. analyzed the tool in a project management context [14]. According to Sheu et al. [15], there are seven main pillars of TRIZ, including the four pillars presented by Mann [16]. In this paper, we will limit the scope to only include the four original pillars presented by Mann:

- Ideality** – A measure of how close a system is to its ideal state. The best value of ideality is infinity. In a project management context, maximizing ideality may involve developing project solutions that generate value for the customer and achieve all project objectives (such as scope, schedule, and budget) with minimal resources, risks, or adverse impacts on stakeholders. Ideality is defined and measured as [15]:

$$\text{Ideality} = \text{Benefits} / (\text{Costs} + \text{Harms}),$$

$$\text{Ideality} = (\text{Perceived Benefits}) / (\text{Costs} + \text{Harms}), \text{ or}$$

$$\text{Ideality} = (\text{Useful Functions}) / (\text{Harmful Functions}).$$
- Contradictions** – Technical contradictions such as trade-offs wherein achieving a desired state is hindered by another aspect of the system that prevents it (improvement in one aspect leads to deteriorations in another). Physical contradictions in which an object or system faces conflicting and opposing requirements [16]. Technical contradictions primarily relate to conflicts within a project's technical parameters and objectives, while physical contradictions involve broader considerations influenced by a project's external physical environment, regulatory frameworks, and stakeholder interests.
- Use of resources** – Optimizing resource utilization within a system is a fundamental principle of TRIZ. In TRIZ terminology, a resource refers to any component within a system that is not fully exploited. This concept encompasses two main strategies: Useless-to-Useful, which involves harnessing resources that are underutilized, discarded, or seemingly irrelevant to the intended purpose, and Harm-to-Help, which focuses on transforming harmful elements into beneficial resources [15]. In project management, using resources in TRIZ may involve identifying underutilized project assets or capabilities and leveraging them to improve project efficiency, innovation, or competitiveness.
- Functionality-Value** – The idea that the system possesses a Main Useful Function (MUF) where any component not contributing to this function is considered harmful (or, in Lean terms, *waste*). In TRIZ, knowledge is classified and arranged by function rather than solution. “Solutions change, functions stay the same.” [9]. Sheu et al. argue that it is really the value we want to achieve with the functionality [15]. In project management, functionality simplification in TRIZ may involve streamlining project workflows, clarifying project requirements, or standardizing processes to improve project clarity, reduce risks, and accelerate project delivery.

Ilevbare [12] discusses the challenges of TRIZ in practice and highlights the following problems to its full application and practitioners' understanding:

- Complexity of the methodology – it is considered too rigid and difficult to adapt. It is considered time-consuming to get a sufficient understanding of the methodology to use it effectively.
- Underlying organizational or cultural issues – adoption of TRIZ is faced with skepticism and fear of change. Lack of a unified standard used by both Russian and Western TRIZ communities.
- Lack of a standard – there is an absence of a best-practice guide for the methodology.

In this paper, we will provide the general background of TRIZ and discuss only the TRIZ tool “40 Inventive Principles” in the context of opportunity management and Lean Construction. By doing this, we are avoiding the issues that are mentioned above and limiting the scope to a part of TRIZ that is not rigid or difficult to adapt. The method is, in many ways, “over-engineered” and challenging to understand, and it will require great commitment and enthusiasm to learn everything. In the research by Ilevbare, 40 Inventive Principles was the most used and best-understood method [12]. Thus, that is considered the method to focus on.

When it comes to the use of TRIZ in the construction industry, research suggests that efforts have been made to introduce TRIZ to construction as a tool for innovation. However, the research on the field is small, and successful applications have depended on creative adaptation [17]. Regarding TRIZ and Lean Construction, one research effort trying to tie the two together is a study from Tommelein and Demirkesen, who looks at TRIZ as a supportive tool for mistake-proofing [18]. Another study from Cabrera and Li evaluates Lean and TRIZ as two complementary principles for improving project resource efficiency [19].

3.2. Opportunity Management and Lean Construction

Opportunity management is about managing the positive effects that emerge during a project and is a part of a project's uncertainty management (the negative effects are referred to as *risks*). Klakegg [20] contends that despite the theory's recognition of risks and opportunities as equally significant, many opportunities are frequently overlooked. Given the substantial uncertainty inherent in projects, effective uncertainty management becomes crucial. The primary objective of Lean Construction can be characterized as creating value for the customer while minimizing non-value-adding activities, commonly referred to as waste [21, 22]. There seems to be a connection between opportunity management and Lean Construction. Torp and colleagues [23] argue that within the context of Lean Construction, the opportunity perspective can be framed as either reducing waste or enhancing value. Malvik et al. evaluate opportunity management and Target Value Delivery (a Lean Construction concept) as complementary practices [7]. Klakegg et al. argue that there is unexploited potential for uncertainty management within Lean Construction [24].

4. TRIZ in a project management context

To show how TRIZ might be useful in a project management context, the TRIZ pillars that were described in section 3.1. are presented with illustrative examples from project management in Table 1.

Table 1. The seven pillars of TRIZ and examples in project management

TRIZ pillars	Project management example
Ideality	Exploit opportunities. Pursue continuous improvement. Minimize waste. Strive for improvements to achieve the ideal project result.
Contradictions	
Technical contradiction	The iron triangle, i.e., increasing quality, often means higher costs or longer project duration. Another example is flexibility vs. stability.
Physical contradiction	For example, if a product is required to be both lightweight and durable. Safety vs. efficiency and security vs. accessibility are other potential contradictions.
Use of resources	
Useless-to-Useful	An area without any function can be transformed into something that generates value and improves the end product. Similarly, subcontractors awaiting their turn can be

	utilized productively during their idle time.
Harm-to-Help	Employee skill development if certain team members lack specific skills. Recycle waste instead of disposing of it.
Functionality-Value	
Main Useful Function (MUF)	Eliminate everything that is not serving the system's MUF. In Lean terms, this will be the same as removing waste and generating value.
Classify by function/value	Provide the correct function to fulfill customer needs. In the early stages of the project, choose a concept that supports the preferred function, not the product.

Table 2 demonstrates the application of six out of the 40 Inventive Principles, showcasing their utility in scenarios aimed at seizing opportunities, minimizing waste, or enhancing value within a project. This table underscores the synergy between TRIZ, opportunity management, and Lean Construction, enabling practitioners to cultivate greater creativity in problem-solving by leveraging TRIZ principles.

Table 2. Various TRIZ Inventive Principles in a Lean Construction and Opportunity Management context.

TRIZ – 40 Inventive Principles	Opportunity Management	Lean Construction
Principle 1 – Segmentation	Break down the project objectives	Single-purpose networks of commitment
Increase the degree of fragmentation or segmentation.	Break down objectives from societal to project performance to secure alignment.	Secure commitment from all project actors to work towards the same objectives.
Principle 2 – Taking out	Evaluate if pre-defined quality aspects are necessary	Remove waste
Separate an interfering part or property from an object, or single out the only necessary part (or property) of an object.	Avoid “gold-plating”.	Remove all non-value-adding activities or aspects.
Principle 3 – Local quality	Improve a part of the building that is not offering anything valuable	Value generation
Make each part of an object fulfill a different and useful function.	Can a gray and boring wall be transformed into a fun climbing/bouldering wall? Can a small area without any function be changed into a small outdoor gym?	Small and simple improvements can generate a lot of value for the customer. However, they will be a wasteful activity if they do not provide value.
Principle 22 – “Blessing in disguise”	Transform adversity into opportunity	Flow and value generation
Use harmful factors (particularly harmful effects on the environment or surroundings) to achieve a positive effect.	When encountering protected or regulated environmental or ecological elements, incorporate them seamlessly into the final product rather than resisting them.	Turn waste into value. Turn wasteful actions into value-generating actions.
Principle 23 – Feedback	User involvement	Continuous improvement
Introduce feedback (referring back, cross-checking) to improve a process or action.	Communicate continuously with the user or customer to understand what they actually value in the final product.	Secure continuous improvement by coupling learning with action.
Principle 26 – Copying	Use a digital twin	Collaboration
Instead of an unavailable, expensive, fragile object, use simpler and inexpensive objects.	Use Building Information Modeling (BIM) to make decisions before it gets too expensive.	Interdisciplinary collaboration with the use of BIM will help practitioners make the correct decision.

Table 2 shows that the Inventive Principles can be applied to both opportunity management and Lean Construction. In this example, we have chosen an overarching project management approach to show the benefits of TRIZ and illustrate the connections between TRIZ, opportunity management, and Lean Construction. In this example, the matter of *project success* has been important. However, since TRIZ was originally developed based on technology patents, its applicability might be even more prominent when the technical aspects of a project are discussed. The TRIZ methodology might be very useful in an uncertainty management situation when considering and deciding on materials and solutions and, thus, help a project achieve *project management success*.

5. Discussion and conclusion

This study has elucidated the potential benefits of integrating the Theory of Inventive Problem-Solving (TRIZ) into opportunity management and Lean Construction methodologies. The TRIZ principles and their application in fostering innovation and creativity can help practitioners identify opportunities, reduce waste, and enhance value within projects. Moreover, the synergies between TRIZ, opportunity management, and Lean Construction highlight their collective role in enhancing project success and long-term sustainability.

The study sought to answer the research question, “How can TRIZ work as a tool to enhance innovation efforts in opportunity management and Lean Construction?”. This was answered by taking a conceptual approach with the aim of analyzing and explaining a predicted relationship between different constructs – that being TRIZ, Lean Construction, and opportunity management – through a literature review. A drawback of TRIZ is its complex nature. Because of this, this research was limited to the four pillars of TRIZ and the tool 40 Inventive Principles. The results show clear connections between TRIZ theory and tools and what you want to achieve with Lean Construction and opportunity management. Opportunities in projects are often more neglected than their counterparts, the threats. With TRIZ, a project organization has a problem-solving tool to help find opportunities when conducting uncertainty management. Adopting TRIZ principles could lead to a shift in mindset among project practitioners, encouraging a more proactive approach to identifying and capitalizing on opportunities.

Further research and practical implementation of TRIZ within project management can contribute to continuous improvement and innovation in construction projects, ultimately improving project outcomes and stakeholder satisfaction. A challenge to implementing TRIZ in practice might be that it was originally intended for technical innovation and requires some mindset adjustments to be applicable in project management. This paper and others have suggested some examples that might help this process, but more can still be done to make it less of a barrier. Also, TRIZ is a complex system, so to reduce the risk of making the implementation too complex, it is suggested to start with the 40 Inventive Principles, which has proved to be the most straightforward tool.

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